

NeuroAnalyzer report: LFP / ERP Analysis

NeuroAnalyzer v0.3.0 · MATLAB 2026a · 2026-09-25T17:59:47+00:00 · window image: exportapp

LFP / ERP Analysis

Load LFP average evoked responses (ERP), map current source density (CSD) and analyze oscillations

? Help

5 Export

Export ERP / CSD...

Save session...

Open session...

Report (PDF)...

Methods.txt...

6 Time-frequency

Try resolution demo

Channel

12x1 (10111)

Frequencies (Hz)

2

00

Wavelet cycles

7

epoch (s)

0.5

1

Baseline (s)

-0.4

-0.1

Band limits (Hz) are conventions of clinical physiology, not filters as needed. Plot = show in band power.

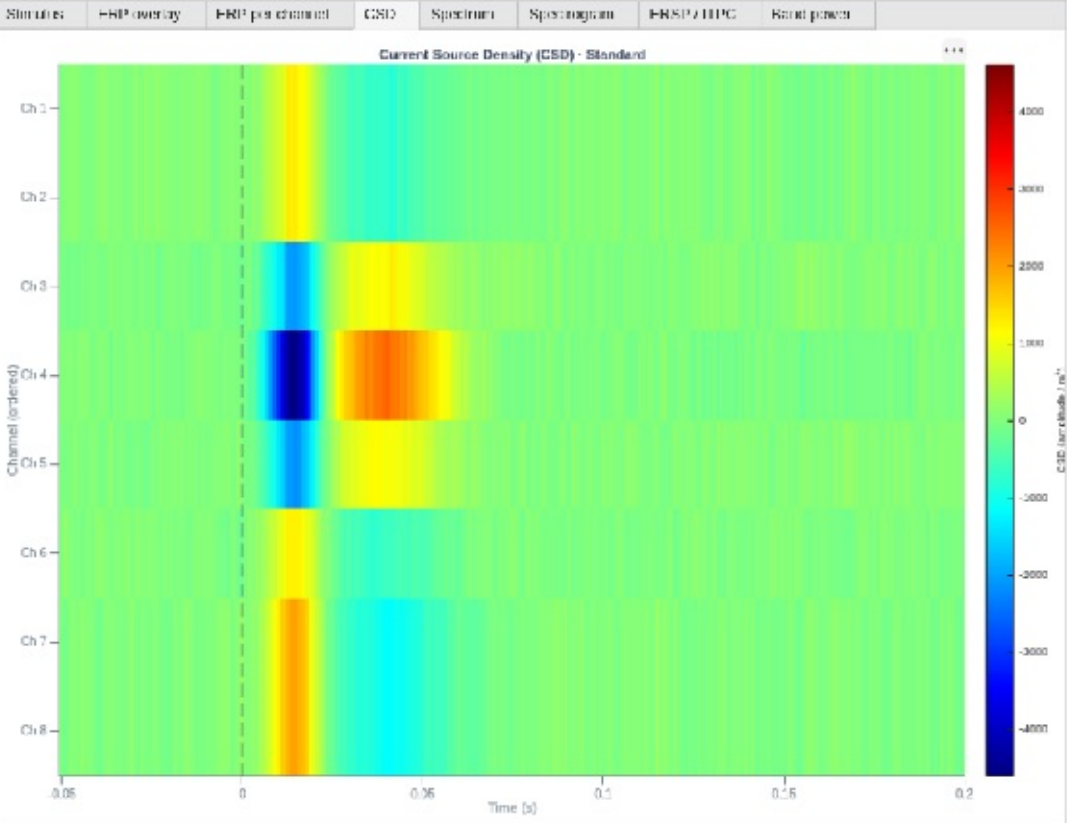
Band	Low (Hz)	High (Hz)	P
Delta	1	4	
Theta	4	8	
Alpha	8	13	
Beta	13	30	
Gamma	30	80	

Spectrum

Spectrogram

ERP / MIPG

Band power



✓ Screenshot of LFPAnalysisApp session mat (1 input file) with MIPG, settings and results.

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Window: LFP / ERP Analysis (LFPAnalysisApp)
NeuroAnalyzer v0.3.0
MATLAB 26.1.0.3346908 (R2026a) Update 5
OS: Linux 6.17.0-1022-azure (GLNXA64)
Session created: 2026-09-25T17:59:47+00:00
Report created: 2026-09-25T17:59:47+00:00

NOTES

LFPAnalysisApp walkthrough: demo data

INPUTS (1)

1. LFP file: demo_lfp.mat
path: /tmp/NeuroAnalyzerDemo/demo_lfp.mat
2,281,464 bytes, modified 2026-09-25T17:54:37
MD5 d27e95a9788939a15f3a3cd039ec0bdb

SETTINGS

```
channels = [1 2 3 4 5 6 7 8]
erp.params.preTime = 0.05
erp.params.postTime = 0.2
erp.params.threshold = 0.5
erp.params.minISI = 0.5
erp.channels = [1 2 3 4 5 6 7 8]
csd.spacingUm = 100
csd.order = 1 2 3 4 5 6 7 8
csd.computed = true
csd.usedOrder = [1 2 3 4 5 6 7 8]
csd.usedSpacingUm = 100
csd.method = standard
csd.params.sigma = 0.3
csd.params.diameterUm = 500
csd.params.smoothUm = 0
csd.params.RUm = 0
csd.params.lambda = 0
csd.usedMethod = standard
csd.usedParams.sigma = 0.3
csd.usedParams.diameterUm = 500
csd.usedParams.smoothUm = 0
csd.usedParams.RUm = 0
csd.usedParams.lambda = 0
tf.channel = 1
tf.fRange = [2 80]
tf.cycles = 7
tf.epoch = [-0.5 1]
tf.baseline = [-0.4 -0.1]
tf.bandNames = {Delta, Theta, Alpha, Beta, Gamma}
tf.bandRanges = [1 4;4 8;8 13;13 30;30 80]
tf.bandPlot = [true true true true true]
tf.focus = false
tfRuns.spectrum = []
tfRuns.spectrogram = []
tfRuns.ersp = []
tfRuns.bandpower = []
tab = CSD
```

KEY RESULTS

LFP: 8 channels at 1017.25 Hz, 30.0 s
ERP: 15 of 15 epochs, channels [1 2 3 4 5 6 7 8], -0.05 to +0.2 s
ERP minimum per channel: [-1.76e-05 -4.8e-05 -9.18e-05 -0.000115 ... 4 more line(s): see the session file